

Entry Name: **JOOS**
VAST Challenge 2025
Challenge 3

Team Members:

Lucas Joos, University of Constance, lucas.joos@uni-konstanz.de PRIMARY
Willi Kneer, University of Constance, willi.kneer@uni-konstanz.de
Jonathan Rentschler, University of Constance, jonathan.rentschler@uni-konstanz.de

Student Team: Yes

Tools Used:

Docker, Gitlab, Kubernetes
Frontend: Next JS + React
Backend: FastAPI, Ollama, Python , Jupyter
Database: Neo4J

Approximately how many hours were spent working on this submission in total?

360 hours

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2025 is complete? Yes

Video

<https://cloud.uni-konstanz.de/index.php/s/a4jWKXSfPsSHZRj>

Use visual analytics to analyze the available data and develop responses to the questions to be provided. In addition, prepare a video that shows how you used visual analytics to solve this challenge.

Note: the VAST challenge is focused on visual analytics and graphical figures should be included with your response to each question. Please include a reasonable number of figures for each question (no more than about 6) and keep written responses as brief as possible (around 250 words per question).

Questions

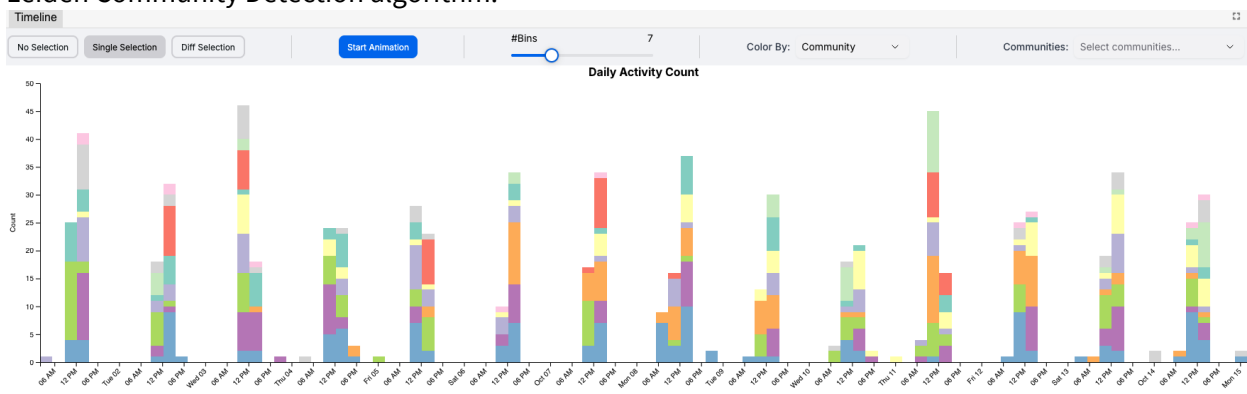
Clepper diligently recorded all intercepted radio communications over the last two weeks. With the help of his intern, they have analyzed their content to identify important events and relationships between key players. The result is a knowledge graph describing the last two weeks on Oceanus. Clepper and his intern have spent a large amount of time generating this knowledge graph, and they would now like some assistance using it to answer the following questions.

- 1 – Clepper found that messages frequently came in at around the same time each day.
 - a. Develop a graph-based visual analytics approach to identify any daily temporal patterns in communications.
 - b. How do these patterns shift over the two weeks of observations?
 - c. Focus on a specific entity and use this information to determine who has influence over them.

1a) Identifying Daily Temporal Patterns

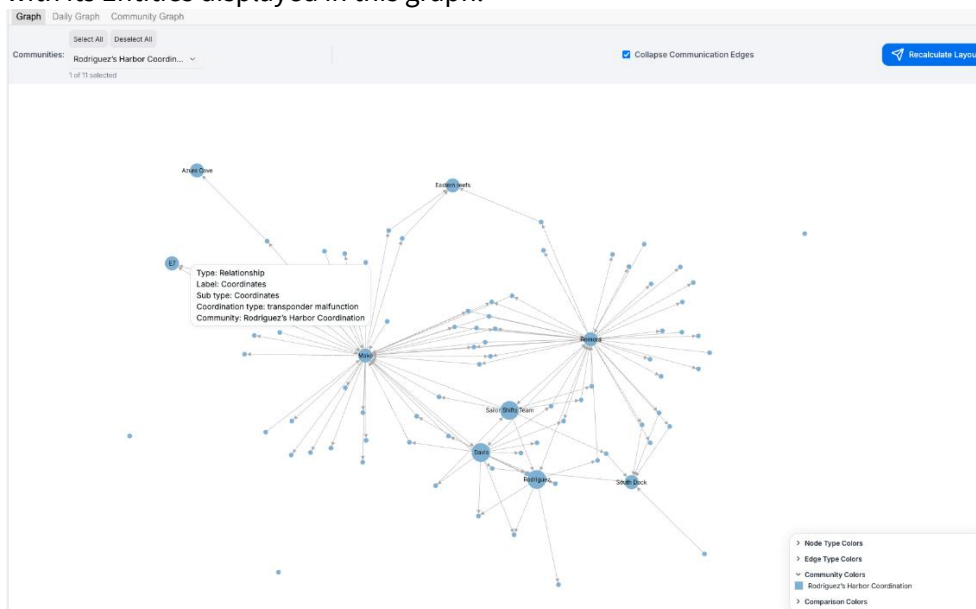
To identify daily temporal patterns in communications, we used a timeline component with brushing, linked to the knowledge graph. This allows us to select a time range and immediately see which events and entities are active during that period.

The following visualization shows the overall temporal pattern by community, as detected using the Leiden Community Detection algorithm:

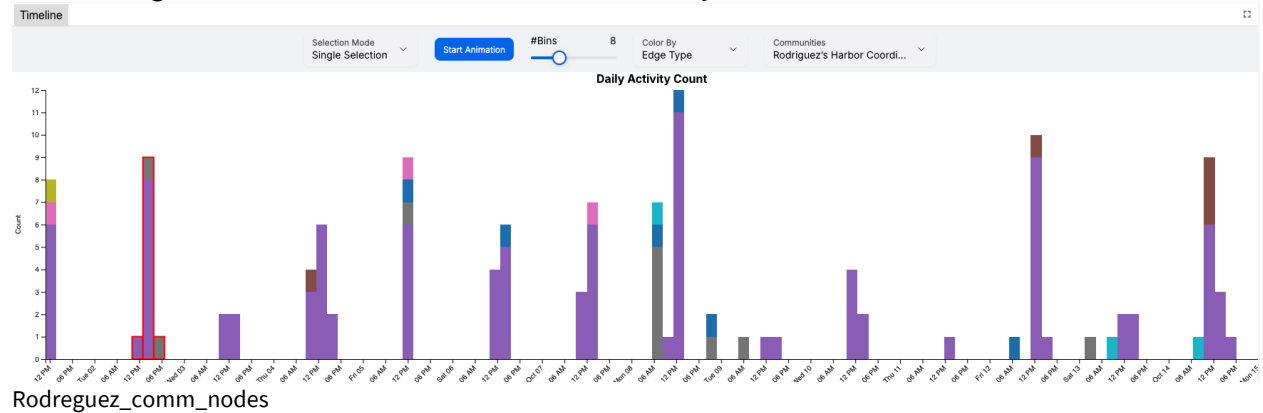


From this visualization we observe that most of the events take place between 10:00 AM and 04:00 PM, distributed between the different communities. There is a consistent peak in activity between 10:00 AM and 12:00 PM each day.

The community, which is present everyday is the Rodriguez Harbor Coordination (Blue Community) with its Entities displayed in this graph:

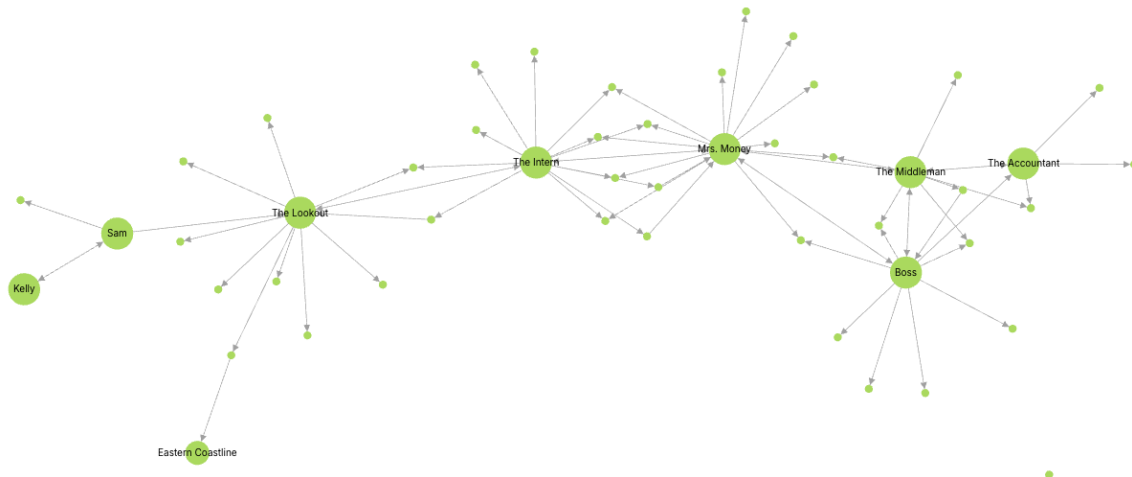


When taking a closer look at the events of the Community:



It is visible that they are mostly communicating (purple bars), with some vessel activity, enforcements, assessments, collaborations and tour activities in between.

The second most present community is the Nemo Reef Operations community (Light Green Community), which has events always around 12AM, except on 06.10.



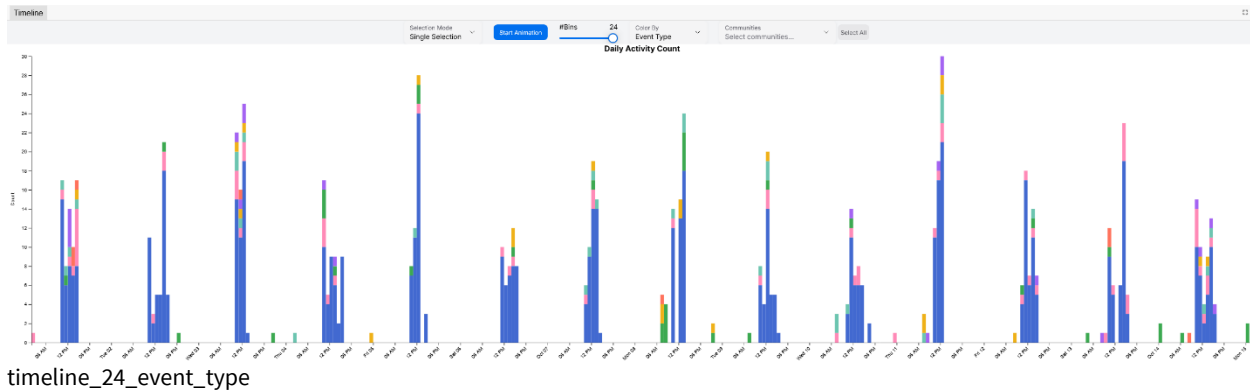
Nemo_Reef_operations_community

1b) Shifts in Patterns Over Two Weeks

Examining the temporal patterns over the two-week period reveals the following:

Intercepted communications consistently occur daily from 11:00 AM to 04:00 PM, including both weekdays and weekends.

There is noticeably less activity during the first weekend (October 6–7) compared to the second weekend (October 13–14).



This suggests that while communication is generally steady, there are fluctuations in activity levels, particularly on weekends.

1c) Entity Focus:

Input:

Query:

Focus on the entity "the Middleman."

Tasks:

1. Identify all individuals, organizations, or entities that have direct or indirect influence over "the Intern."
2. Determine the types of relationships or interactions that indicate influence (e.g., mentorship, supervision, information control, resource allocation).
3. Summarize the extent and nature of each influencing party's control or leverage over "the Middleman."

Output:

2 – Clepper has noticed that people often communicate with (or about) the same people or vessels, and that grouping them together may help with the investigation.

- a. Use visual analytics to help Clepper understand and explore the interactions and relationships between vessels and people in the knowledge graph.
- b. Are there groups that are more closely associated? If so, what are the topic areas that are predominant for each group?
 - For example, these groupings could be related to: Environmentalism (known associates of Green Guardians), Sailor Shift, and fishing/leisure vessels.

2a) Visual Analytics for Exploring Interactions and Relationships

To help Clepper explore the interactions and relationships between vessels and people, we applied the Leiden algorithm to detect communities within the knowledge graph. This approach revealed eleven distinct communities, with each node assigned to a specific group.

In our Graph RAG pipeline, these communities are automatically detected and summarized using a large language model. The summaries are accessible by hovering over nodes in the `Community Graph` visualization, providing instant context for each group.

The identified communities are as follows:

1. Nemo Reef Protection
2. Himmap & Dolphin Bay Management
3. Rodriguez's Harbor Coordination
4. Nemo Reef Operations
5. Restricted Access Issues
6. Nemo Reef Monitoring by EcoVigil
7. Nemo Reef operations by V. Miesel
8. Access Management
9. Nemo Reef interactions by Oceanus City
10. Nemo Reef Unauthorized Activity
11. Nemo Reef & Haacklee Harbor

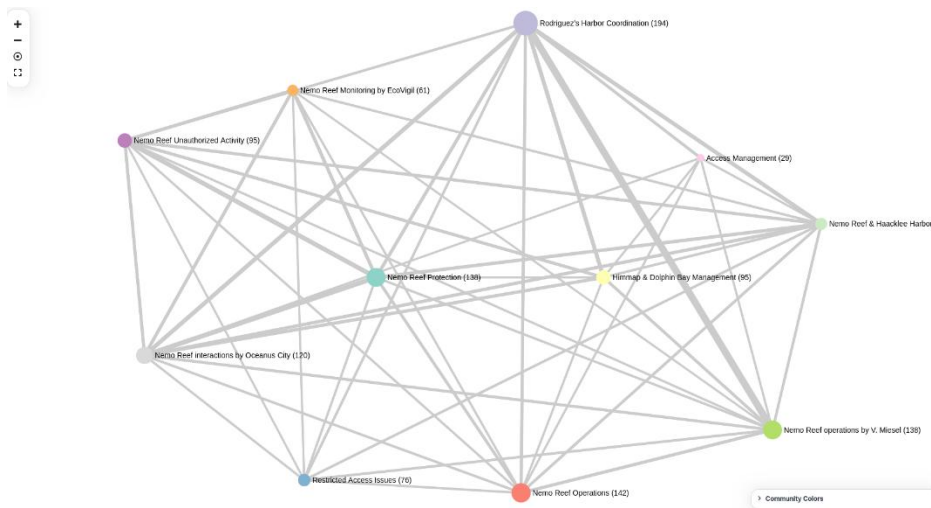


Figure: Community Graph Visualization

With the community data, we can change the coloring of the Timeline as well as the Node Link diagram of the knowledge graph. To help the analysis of specific communities, we can filter the nodes based on the community in the Node Link diagram and.

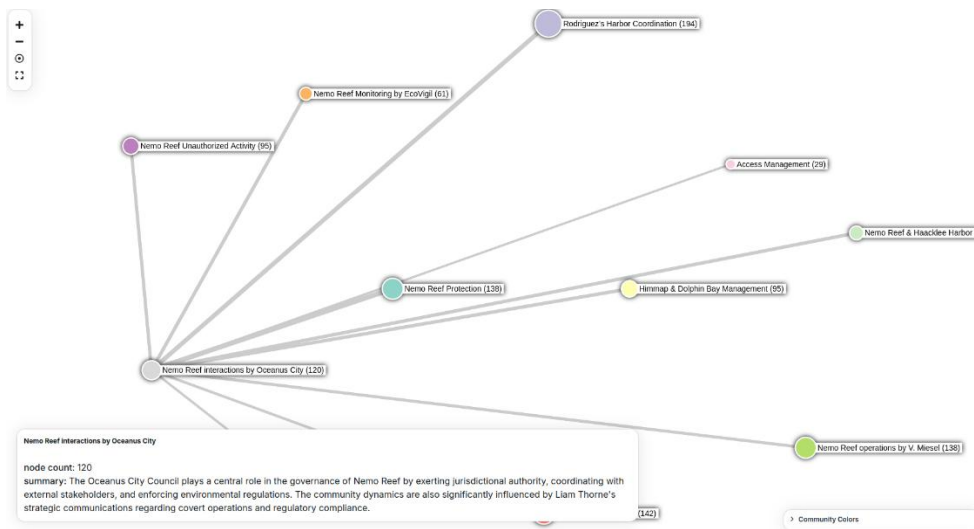


Figure: Community Graph visualization showing the "Nemo Reef interactions by Oceanus City" community highlighted.

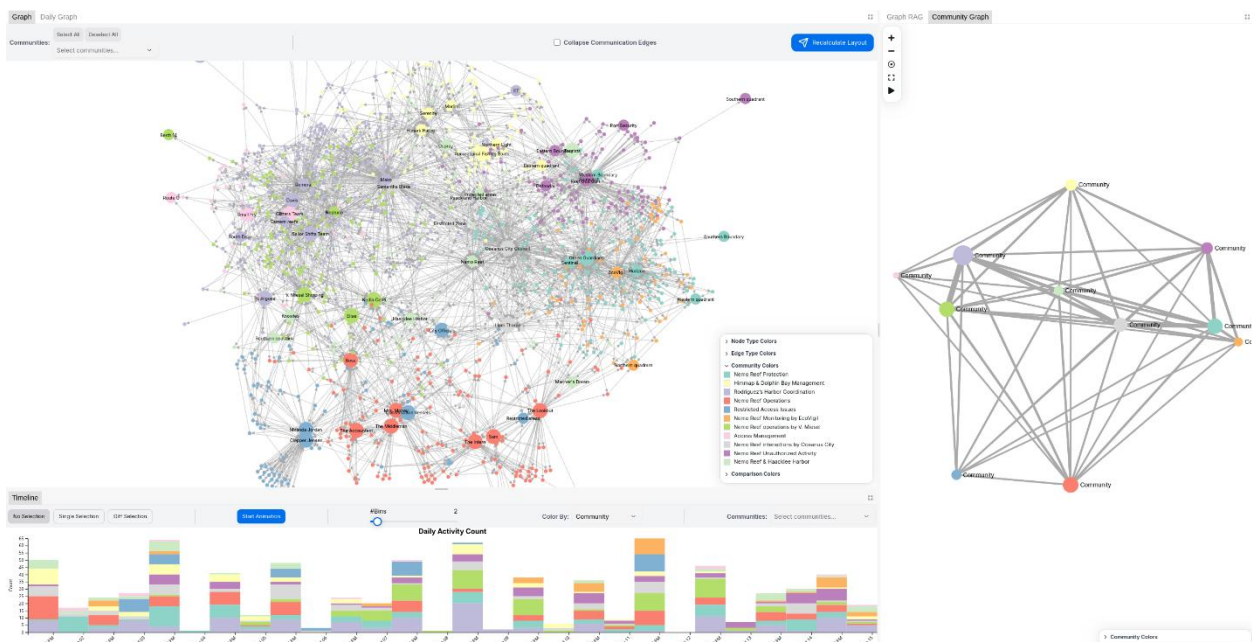


Figure: Knowledge Graph with the coloring based on the community (top) in timeline of events colored by the community (bottom).

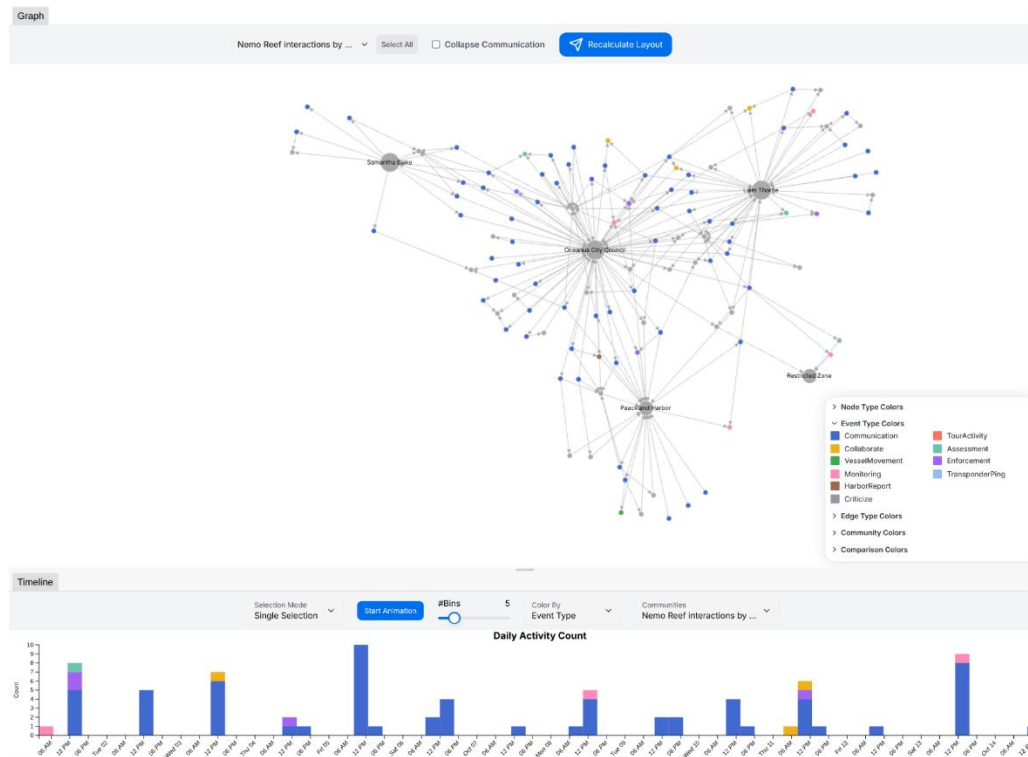


Figure: Knowledge Graph with the community `Nemo Reef interactions by Oceanus City` selected (top) and the timeline of events associated with the same community (bottom)._

2b) Community Associations and Predominant Topics

The communities "Nemo Reef Operations by V. Miesel" and "Rodriguez's Harbor Coordination" are highly interconnected, indicating frequent interactions and possibly shared interests or collaborations. The edge size in the community graph is proportional to the number of interactions between the two communities. For these two communities, the edge size is relatively large.

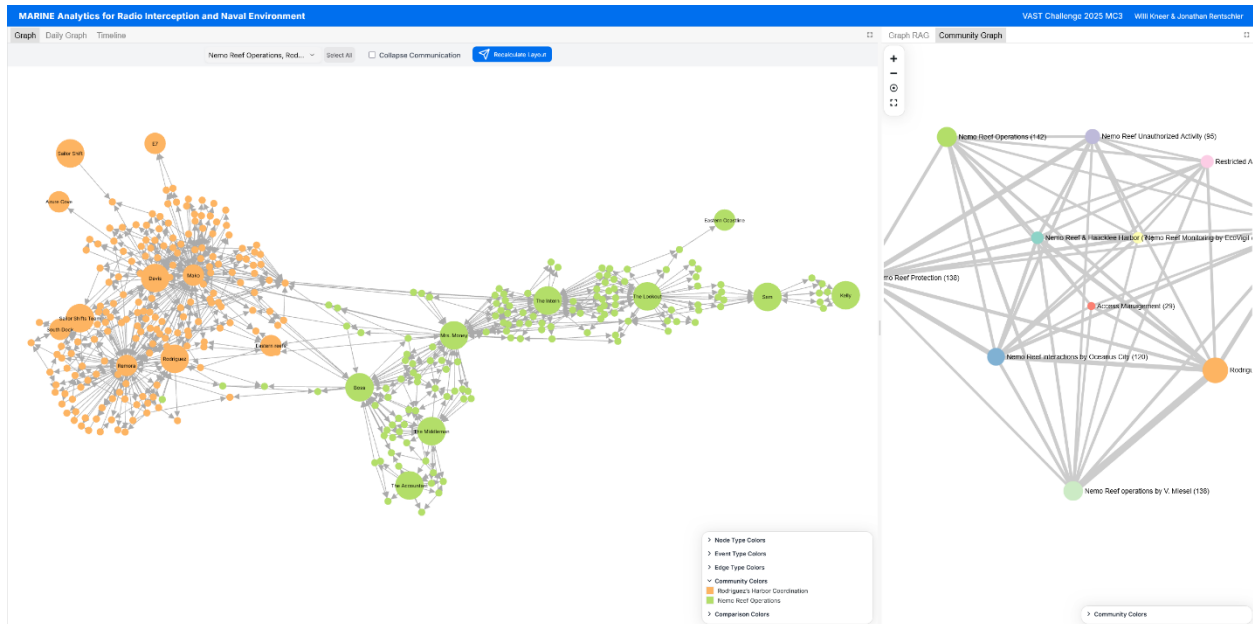


Figure: Community graph with the two communities "Nemo Reef Operations by V. Miesel" and "Rodriguez's Harbor Coordination" highlighted.

Approximately half of the identified communities are primarily concerned with issues related to Nemo Reef. This highlights that the Nemo Reef is a central topic in the network.

- 3 – It was noted by Clepper's intern that some people and vessels are using pseudonyms to communicate.
 - a. Expanding upon your prior visual analytics, determine who is using pseudonyms to communicate, and what these pseudonyms are.
 - Some that Clepper has already identified include: “Boss”, and “The Lookout”, but there appear to be many more.
 - To complicate the matter, pseudonyms may be used by multiple people or vessels.
 - b. Describe how your visualizations make it easier for Clepper to identify common entities in the knowledge graph.
 - c. How does your understanding of activities change given your understanding of pseudonyms?

3a) Pseudonym Detection

After looking over the Knowledge Graph on Community of Persons and Names that looks very suspicious:

1. The Accountant (Davis aber nicht save)
2. Boss (Nadia Conti)
3. Small Fry (Rodriguez)
4. The Middleman (Liam Throne)

5. Mrs. Money (Elise)
6. The Intern (Sam)
7. The Lookout (Kelly)

After asking the GraphRAG Pipeline what this Group is all about, we got this Answer:

I found that there is a group of persons, which are connecting a lot including: Boss, the Intern, the Middleman, The accountant, and Mrs. Money. What are they doing and who are they?

The group consisting of Boss, the Intern, the Middleman, the Accountant, and Mrs. Money are involved in a clandestine operation with activities spanning financial manipulation, potential environmental exploitation, and attempts to evade detection ([Subgraph 0 Subgraph 2 Subgraph 4](#)).

Roles and Responsibilities:

Boss appears to be the leader or primary decision-maker ([Subgraph 0 Subgraph 4](#)). Mrs. Money is a key operative responsible for financial management, securing alternative funding (through V. Miesel Shipping), erasing financial trails, and implementing contingency protocols ([Subgraph 0 Subgraph 2 Subgraph 4](#)). The Middleman acts as an intermediary with connections to Council ([Subgraph 0 Subgraph 4](#)), and redirects Council's attention away from Nemo Reef ([Subgraph 4](#)). The Accountant handles financial records and is potentially involved in concealing financial activity ([Subgraph 0](#)). The Intern assists Mrs. Money, compiles reports, and carries out tasks as assigned ([Subgraph 2 Subgraph 3](#)).

Activities:

The group's activities initially centered on exploiting a loophole in protection zones around Nemo Reef, disguising this under the guise of tourism ventures ([Subgraph 2](#)). They actively conceal financial trails related to Nemo Reef ([Subgraph 0 Subgraph 1 Subgraph 2](#)), manage Council by redirecting their attention ([Subgraph 4](#)), and frequently have meetings to coordinate strategy ([Subgraph 0 Subgraph 1 Subgraph 4](#)). Earlier plans were halted due to increased scrutiny, leading to a redirection of funds through V. Miesel Shipping ([Subgraph 2](#)).

Show Graph

Knowledge RAG Input

Enter a Question to build a Knowledge Graph or get Information about Person, Vessles or Locations



Submit



Context Search

Figure: Prompt "I found that there is a group of persons, which are connecting a lot including: Boss, the Intern, the Middleman, The accountant, and Mrs. Money. What are they doing and who are they?"

The Middleman

Looking at this Information we found that the easiest Entity to uncover would be the Middleman. Because he is a Person talking to the City council, who has specific information only he and the Middleman could have.

So we selected the Community with the City Council in the Bar chart to look at the events they have.

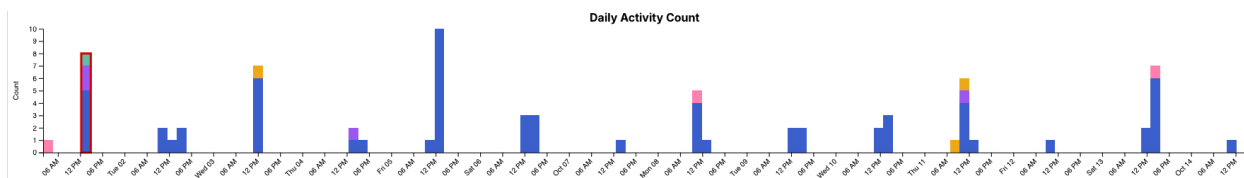


Figure: City Council Events Timeline

And when looking at the events, we also included the Community with all the discovery nick names. Which led to this graph for the first time-stamp (01.10 10:00 AM - 01:00 PM).

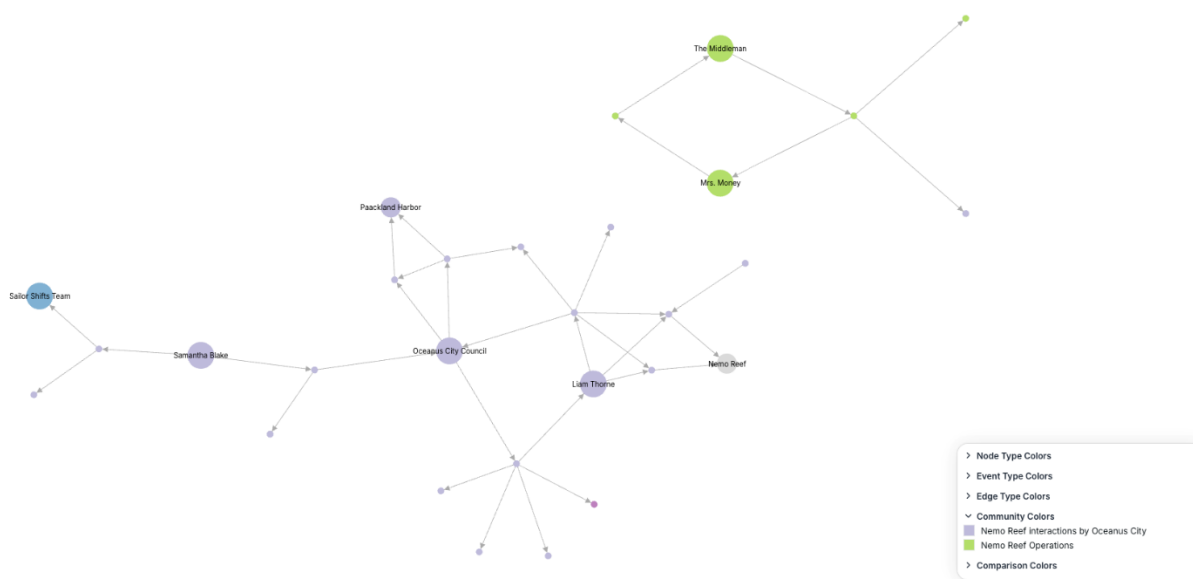


Figure: Liam Thornes @ 01.10 10:00 AM - 01:00 PM

Here at 10:51 AM the City Council asks Liam to increase the patrols at Nemo Reef, because they heard about increased vessel activity there. At 10:53 AM the Middleman sends to Mrs. Money that the Council has flagged increased activity at Nemo Reef.

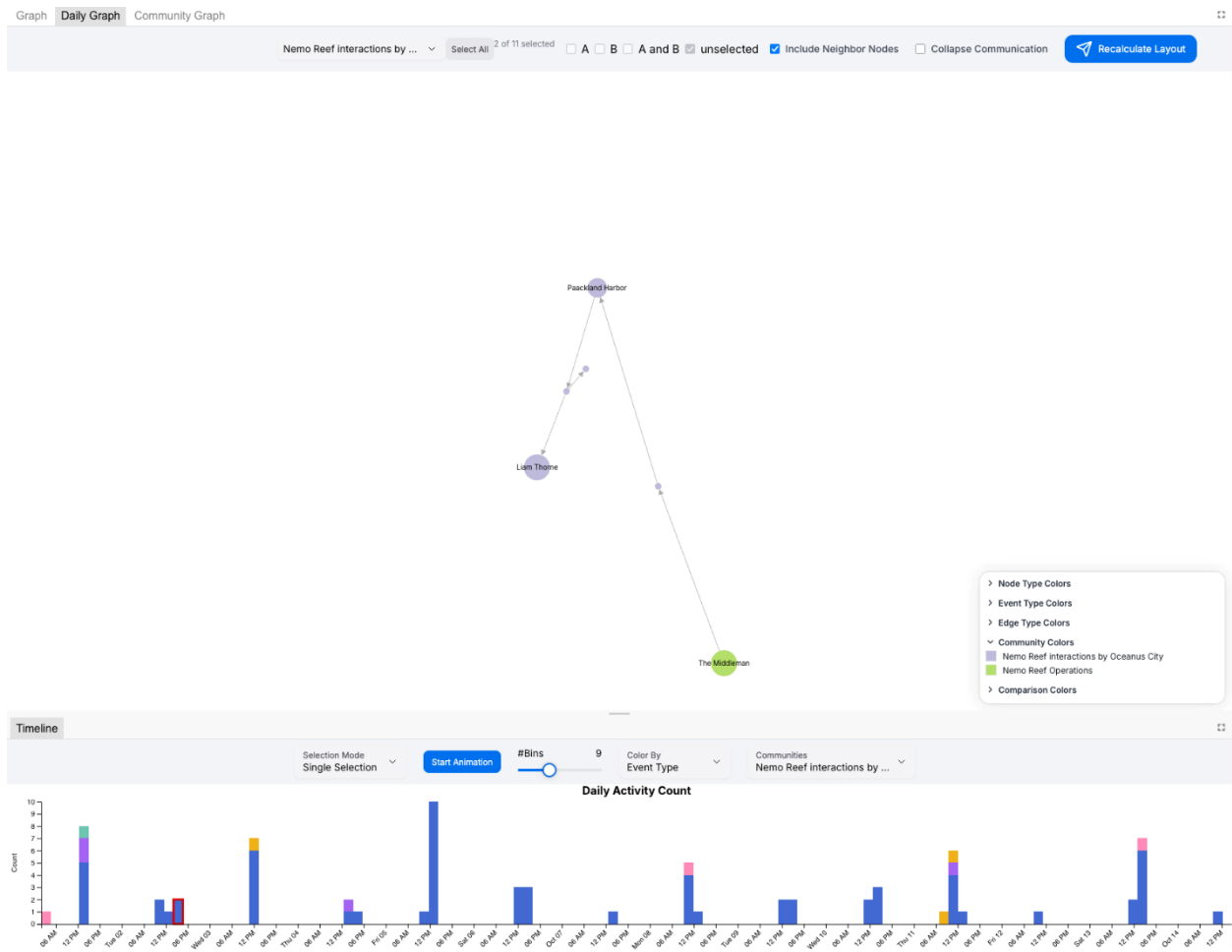


Figure: Liam Thornes as Middleman

And here the Paackland Harbour asks Liam for a report, and he answers as the Middleman. Fully exposing himself.

Mrs. Money

The next easiest Person to uncover was Mrs. Money. While reading the GraphRAG Answer, we discovered that Mrs. Money must be a Person who often communicates with V. Miesel Shipping. And after taking the same approach as for the Middleman, we found out that Mrs. Money is Ellise.

The Intern and the Lookout

While investigating patterns in the graph, we found that Kelly and Sam's communication and the Lookout's and the Interns' communication are about the same things. With Kelly being the Lookout and Sam being the Intern.

Small Fry

With the GraphRAG we found out that Rodriguez is Small Fry.

3b) Common Entities in the Knowledge Graph

3c) Understanding Activities with Pseudonyms

4 – Clepper suspects that Nadia Conti, who was formerly entangled in an illegal fishing scheme, may have continued illicit activity within Oceanus.

- a. Through visual analytics, provide evidence that Nadia is, or is not, doing something illegal.
- b. Summarize Nadia's actions visually. Are Clepper's suspicions justified?

4a) Evidence of Illegal Activity

4b) Summarizing Nadia's Actions

Reflection Questions (Optional)

- Given the task to develop visualizations for knowledge graphs, did you find that the challenge pushed you to develop new techniques for visual representation?
- Did you participate in last year's challenge? If so, did your experience last year help prepare you for this year's challenge?
- What was the most difficult part of working on this year's data and what could have made it more accessible?

We did not participate in last year's challenge.

Working with the html Form was a bit difficult, because not all our machines used supported Microsoft Word. Markdown or latex would have been easier to work with.

All VAST data is fully synthetic. Any resemblance to real people, places, or events is purely coincidental. Some elements of the 2025 VAST Challenge resemble data released for the 2024 VAST Challenge. Participants should not assume that there is any continuity and should not use any earlier or any external data for their submission. Mini-Challenge 3 participants should only use data supplied for this mini-challenge in their submission. Data from other challenges should not be combined, except in the Design Challenge.